Searching for Dermatology Information Online using Images vs Text: a Randomized Study

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Introduction

Skin conditions are prevalent globally and significantly impact anxiety and morbidity. While individuals commonly use text-based search (e.g., “red rash on arm”) to investigate skin concerns, this process is often hindered by challenges in accurately describing lesion morphology. This study evaluates user experiences with an artificial intelligence image-based search compared to traditional text-based search for identifying skin conditions.

Methods

- An internet-based survey recruited 372 respondents from a commercial survey panel, including participants willing to photograph their visible skin concerns.
- Using the Google mobile app, respondents conducted both text-based (Google Search) and image-based (Google Lens) searches with the order randomized.
- Satisfaction across six dimensions was recorded for each modality.
- Preferences between modalities were compared.

Results

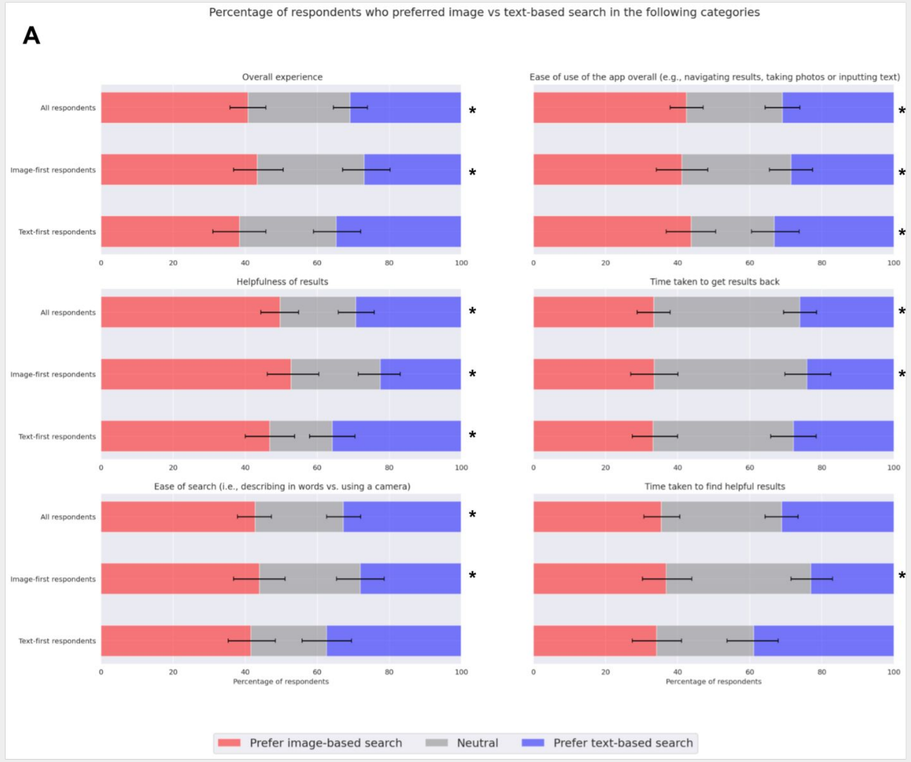
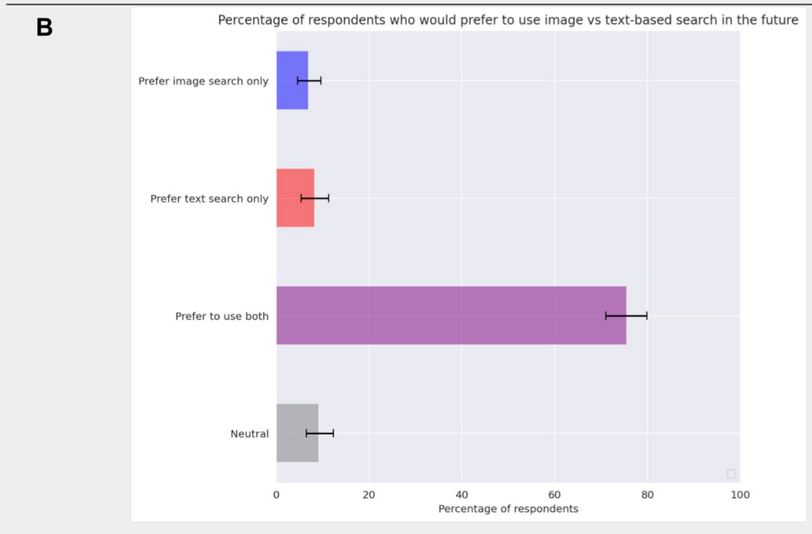
- Of the respondents, 44% identified as women, 86% as White, and 41% were over age 45.
- 81.5% and 63.5% were at least moderately familiar with text-based and image-based search respectively.
- Respondents expressed high satisfaction with both modalities, with over 90% at least somewhat satisfied across all dimensions.
- Direct comparisons revealed a preference for image-based search in 5 of 6 dimensions, with a 9.9% overall preference for image-based search (p=0.004).
- Notably, 82.5% (95% CI 78.2–86.3) expressed a desire to incorporate image-based search in future queries, whether alone or in combination with text-based search, with 64% of these favoring image-based search as the initial approach.

Table 1: Patient Characteristics

Age	18-34: 45.2% 35-54: 29.3% >54: 25.5%
Gender	Woman: 43.5%
Race	White: 85.8%
Ethnicity	Hispanic, 14.0%
Education level	>=Bachelor’s: 61.%
Familiarity with ...	Image-based search: 63.5% Text-based search: 81.5%
Fitzpatrick skin type	1: 11.8% 2: 37.4% 3: 33.9% 4: 8.6% 5: 6.2% 6: 1.6%

Conclusion

- Despite less familiarity, participants favored image-based search for skin conditions and indicated a strong preference to integrate it into future searches.
- These findings highlight the potential of image-based search as a key tool for improving the accessibility and accuracy of online information regarding skin concerns.

References

1. Hay, R. J., Johns, N. E., Williams, H. C., Bolliger, I. W., Dellavalle, R. P., Margolis, D. J., ... & Naghavi, M. (2014). The global burden of skin disease in 2010: an analysis of the prevalence and impact of skin conditions. Journal of investigative dermatology, 134(6), 1527-1534.
2. Ahuja, S., Briggs, S. M., & Collier, S. M. (2022). Teledermatology in rural, underserved, and isolated environments: a review. Current dermatology reports, 11(4), 328-335.
3. Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. nature, 542(7639), 115-118.
4. Jeong, H. K., Park, C., Henao, R., & Kheterpal, M. (2023). Deep learning in dermatology: a systematic review of current approaches, outcomes, and limitations. JID Innovations, 3(1), 100150.